Stochastic Processes Semester 1

Extension Questions 3 – Solutions

1. (a) i. We have

$$P^2 = \begin{pmatrix} 1/3 & 2/9 & 2/9 & 2/9 \\ 1/6 & 5/12 & 1/4 & 1/6 \\ 1/9 & 1/6 & 7/18 & 1/3 \\ 0 & 0 & 0 & 1 \end{pmatrix} \qquad P^3 = \begin{pmatrix} 5/27 & 5/27 & 1/3 & 8/27 \\ 5/36 & 7/24 & 23/72 & 1/4 \\ 1/6 & 23/108 & 17/108 & 25/54 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

ii.

$$\mathbf{P}(X_{74} = 1 \,|\, X_{71} = 3) = 1/6.$$

(b) i. Let $\boldsymbol{\mu}^{(n)} = (\mu_i^{(n)})$ where $\mu_i^{(n)} = \mathbf{P}(X_n = i)$. Then

$$\boldsymbol{\mu}^{(3)} = \begin{pmatrix} 53/324, & 149/648, & 175/648, & 109/324 \end{pmatrix}$$

ii. We further have

$$\mathbf{P}(X_0 = i \,|\, X_3 = 4) = \begin{pmatrix} 32/109, & 27/109, & 50/109, & 0 \end{pmatrix}, \qquad i = 1, 2, 3, 4.$$

2. (c) i. Let

$$\boldsymbol{\mu}^{(n)} = (\mathbf{P}(X_n = 0), \mathbf{P}(X_n = 1), \mathbf{P}(X_n = 2))$$

then

$$\boldsymbol{\mu}^{(n)} = \begin{pmatrix} 1/3 & 1/3 & 1/3 \end{pmatrix} + \left(-\frac{1}{2}\right)^n \begin{pmatrix} 1/6 & 1/6 & -1/3 \end{pmatrix}.$$

ii.

$$\lim_{n\to\infty} \boldsymbol{\mu}^{(n)} = \begin{pmatrix} 1/3 & 1/3 & 1/3 \end{pmatrix}.$$

(d)

$$\boldsymbol{\mu}^{(n)} = \begin{pmatrix} 1/3 & 1/3 & 1/3 \end{pmatrix}$$

i.e. the initial distribution is *stationary*.

3. For general p, the transition matrix is

$$P = \begin{pmatrix} p & 1-p & 0 & 0 \\ p & 0 & 1-p & 0 \\ p^2 & p(1-p) & 0 & 1-p \\ p^3 & p^2(1-p) & p(1-p) & 1-p \end{pmatrix}.$$

(a) When $p = 1/10$ the probability that a policy holder who has the 20% discount has the 60% discount after 4 years is $86022 \times 10^{-5} = 0.86022$

(b) For general p the probability that a policy holder who has the 60% discount has no discount after 2 years is $p^3(3 - 2p)$

4. (a) The classes of intercommunicating states are

Class	Type
$\{1,3,5,7\}$	open
$\{2\}$	closed (absorbing)
$\{4\}$	open
$\{6\}$	closed (absorbing)

The vectors of absorption probabilities associated with the closed classes are

$$\boldsymbol{\alpha}^{\{2\}} = (1,1,1,0,1,0,1)^T$$
$$\text{and } \boldsymbol{\alpha}^{\{6\}} = (0,0,0,1,0,1,0)^T$$

(b) The classes of intercommunicating states are

Class	Type
$\{1\}$	closed (absorbing)
$\{2,3\}$	closed
$\{4,5,7\}$	open
$\{6\}$	open

The vectors of absorption probabilities associated with the closed classes are

$$\boldsymbol{\alpha}^{\{1\}} = (1,0,0,4/7,9/14,25/42,23/42)^T$$
$$\text{and } \boldsymbol{\alpha}^{\{2,3\}} = (0,1,1,3/7,5/14,17/42,19/42)^T$$

(c) The classes of intercommunicating states are

Class	Type
$\{1\}$	closed (absorbing)
$\{2,4,5\}$	open
$\{3\}$	closed (absorbing)
$\{6,7\}$	closed

The vectors of absorption probabilities associated with the closed classes are

$$\boldsymbol{\alpha}^{\{1\}} = (1,5/13,0,6/13,2/13,0,0)^T,$$
$$\boldsymbol{\alpha}^{\{3\}} = (0,6/13,1,2/13,5/13,0,0)^T$$
$$\text{and } \boldsymbol{\alpha}^{\{6,7\}} = (0,2/13,0,5/13,6/13,1,1)^T$$

5. (a) $\{Z_n\}_{n\geq 1}$ is not a Markov Chain. To see this compare

$$\mathbf{P}\left(Z_{n+1}=(2,1)|Z_n=(1,1),Z_{n-1}=(0,1)\right)$$
$$\text{and } \mathbf{P}\left(Z_{n+1}=(2,1)|Z_n=(1,1),Z_{n-1}=(2,1)\right)$$

(b) i. $\{X_n\}_{n\geq 1}$ is not a Markov Chain. To see this compare

$$\mathbf{P}\left(X_{n+1}=1|X_n=0,X_{n-1}=0\right)$$
$$\text{and } \mathbf{P}\left(X_{n+1}=1|X_n=0,X_{n-1}=1\right)$$

ii. $\{Z_n\}_{n\geq 1}$ is not a Markov Chain. To see this compare

$$\mathbf{P}\left(Z_{n+1}=(0,1)\,|Z_n=(0,0)\,,Z_{n-1}=(0,0)\right)$$
$$\text{and } \mathbf{P}\left(Z_{n+1}=(0,1)\,|Z_n=(0,0)\,,Z_{n-1}=(1,0)\right)$$

6. (a) We could have a state space which records, how many umbrellas are at each location and where the academic is, e.g. $(0,2,o),(0,2,h),(1,1,o)$, etc. where "o" means "office" and "h" means "home". But this is too much information for what we are interested in.

   Consider instead the number of umbrellas at the same location as the academic, so $\mathcal{S} = \{0,1,2\}$.

   (b) Clearly, if $X_n = 0$, then $X_{n+1} = 2$ (both umbrellas are at other station, and the academic finds it there upon arrival).
   If $X_n = 2$, then $X_{n+1} = 1$ if raining (with probability p) or $X_{n+1} = 0$ if not (with probability $1-p$).
   Finally, if $X_n = 1$, then $X_{n+1} = 2$ with probability p and $X_{n+1} = 1$, otherwise.
   So the transition matrix is

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1-p & p \\ 1-p & p & 0 \end{pmatrix}$$

   (c) We are interested in probability $\mathbf{P}(X_3 \geq 1 \mid X_0 = 2)$. We have

$$P^2 = \begin{pmatrix} 1-p & p & 0 \\ p(1-p) & 1-2p+2p^2 & p(1-p) \\ 0 & p(1-p) & 1-p+p^2 \end{pmatrix}$$

   so

$$\mathbf{P}(X_3 \geq 1 \mid X_0 = 2) = 1 - \mathbf{P}(X_3 = 0 \mid X_0 = 2) = 1 - p_{20}^{(3)}$$

   where

$$p_{20}^{(3)} = p_{20}p_{00}^{(2)} + p_{21}p_{10}^{(2)} + 0 = 1 - 2p + 2p^2 - p^3.$$

   So

$$\mathbf{P}(X_3 \geq 1 \mid X_0 = 2) = p^3 - 2p^2 + 2p.$$